An Unreasonably Cheap Algorithm to Generate Extreme Weather Events

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Introduction

Critical systems and extreme weather

- Energy, transport, agriculture: weather dependent
  - Short-to-medium-term operation $\leftarrow$ forecasting
  - Long-term planning $\leftarrow$ scenario analysis (including extremes)
- Extremes: unlikely but highly relevant for decision-making
- Analyze consequences $\rightarrow$ map bottlenecks & points of failure

$\rightarrow$ **How can these scenarios be generated?**

High-level objective

Objective: simulate *extreme*, yet *plausible* weather scenarios

- Extreme + plausible == usability requirements
- Scenarios: simulate at different intensity levels
- Example:
  - if the average temperature
  - over the next N days
  - over this region
  - goes up by 2°, 3°, 4°
  - $\rightarrow$ then ...

Definitions:

- *Extreme*
  - unlikely ($\rightarrow$ difficult to predict) and disruptive
  - tails of the climatological and forecasting distribution
- *Plausible*
  - spatial- and pressure-level change
  - variables co-variation

Sub-objectives and building blocks

We want:

- a **practical** algorithm for generating extreme scenarios
- scenarios can be **explicitly specified**
- **low** compute and memory cost

We combine two ingredients:

1. Base model

- *ArchesWeather*
- medium-range generative probabilistic forecasts
- flow matching

2. Guidance methods

- *classifier-free guidance*
- borrowed from image generation domain
- **adapted to** weather domain

Images vs. extreme weather

What and why **adapt**?

$\rightarrow$ 3 key differences:

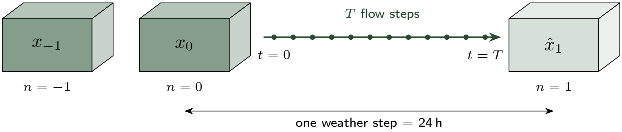
- **Steps:**
  - images $\rightarrow$ final image obtained one-shot
  - weather $\rightarrow$ **multi-step trajectory** (video-alike, but not really ...)
- **Evaluation:**
  - image $\rightarrow$ visual appeal
  - weather $\rightarrow$ qualitative (visual) + quantitative **plausibility** tests
- **Target:**
  - image $\rightarrow$ inherent to a specific task (ground-truth available)
  - weather $\rightarrow$ unclear a priori (**designed**)

Designing guidance

Time

Everything that follows runs on **two nested clocks**:

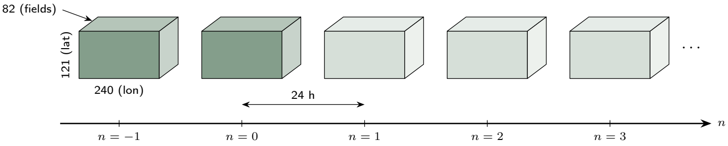
$n = 0, \ldots, N$	**weather time** (data): 24 h forecast steps
$t = 0, \ldots, T$	**flow time** (model): T sampling steps



Data

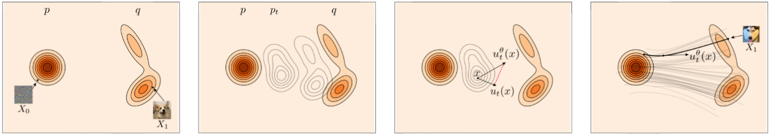
ERA5 Reanalysis (ECMWF)

- Global, 1.5° grid — 240×121 points
- 82 fields per state:
  - 4 surface: 2m temperature, MSL pressure, 10m u/v wind
  - 6 upper-air $\times$ 13 pressure levels: geopotential, temperature, humidity, u/v wind, vertical velocity
- 24-hour forecast steps, available at 00:00, 06:00, 12:00, 18:00
- Test set: $\geq$ 01.01.2020



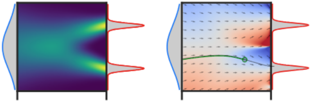
Model

Before introducing the specifics of ArchesWeather ...
... **flow matching** at a glance:



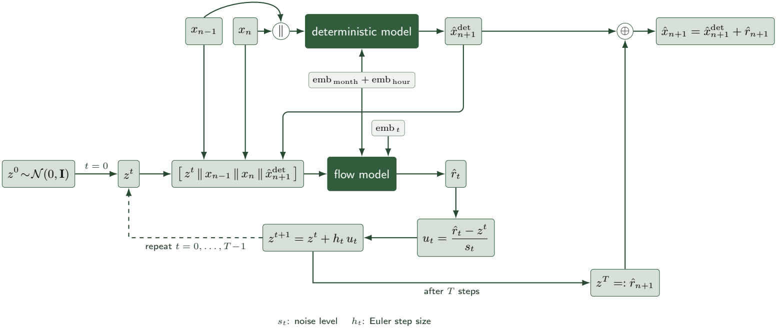
(a) Data. **(b)** Path design. **(c)** Training. **(d)** Sampling.

→ In our work we modify (d):



Model

- Inference: deterministic model + residual flow model
- Flow model: transports $z^0 \sim \mathcal{N}(0, \mathbf{I}) \to z^T \sim p\big(x_{n+1} - \hat{x}^{\text{det}}_{n+1}\big)$
- Note: diffusion is performed in data space, not in latent space.



Target scenarios

We target **heatwave** scenarios over specific:

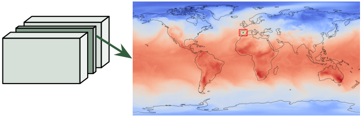
- **mask**: weights over region + variable of interest
- **forcing trajectory**: masked-average percentage change

... what does this mean practically?

- **mask**: localized change of interest, although change will expand beyond mask –> in evaluation we care about weather being consistent over the region of interest, not globally
- **forcing trajectory**: explicit definition of heatwave trajectories, and on how to sample relevant ones. For instance: .... (figure) –> we defer this to future work, here we focus on the guidance method and its evaluation, rather than on the definition of appropriate trajectories

Mask

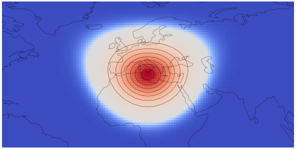
Mask:

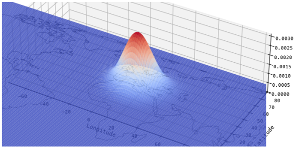


- Place bounding box:
  - over the region of interest
  - at the variable of interest

- Define Gaussian on the sphere:
  - μ = rectangle center; $\sigma_{\text{lat}}, \sigma_{\text{lon}}$ = box sides;
  - $d_{\text{ns}}, d_{\text{ew}}$ = great-circle offset to μ
  - $\cos(\text{lat})$: cell-area weight of equiangular grid

$$\tilde{m} \;=\; \exp\left(-\frac{1}{2}\left[\frac{d_{\text{ns}}^2}{\sigma_{\text{lat}}^2} + \frac{d_{\text{ew}}^2}{\sigma_{\text{lon}}^2}\right]\right)\cos(\text{lat}), \qquad m \;=\; \frac{\tilde{m}}{\sum_{\text{lat,lon}} \tilde{m}}$$

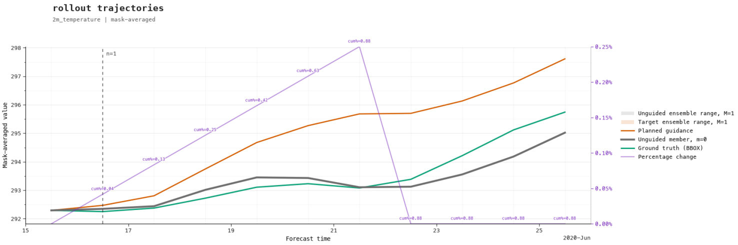




Forcing trajectory

- over an N-step timeframe
- with respect to the unguided step average over the mask (later: on-line w.r.t. guided–unguided step)

TODO: figure doesn't make sense! –> also, we could directly follow the planned trajectory as target, why complicating with the gui-ung step in the first place?



Classifier-free guidance

Steering with a loss gradient $=$ sampling from a posterior

$$p(z^t|y) \propto p(y|z^t)\, p(z^t)$$
$$\nabla_{z^t} \log p(z^t|y) = \nabla_{z^t} \log p(z^t) + \nabla_{z^t} \log p(y|z^t)$$

- define likelihood: $p(y|x) \propto e^{-L(x)}$ where $x := \hat{x}_T$

$$p(y|z^t) = \int p(y|x)\, p(x|z^t)\, dx = \mathbb{E}_{x|z^t}\left[e^{-L(x)}\right] \approx e^{-L(\hat{x}_t)}$$

Guided vector field

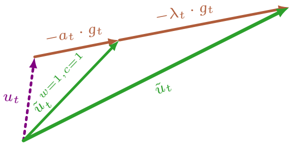
$$\tilde{u}_t = u_t - \lambda_t\, g_t, \quad \text{with } \lambda_t := w \cdot a_t \cdot c_t$$
$$g_t = \nabla_{z^t} L(\hat{x}_t) = -\nabla_{z^t} \log p(y|z^t)$$

- exact posterior needs $w = 1$ and $c_t = 1$ with path-specific a_t (e.g. $\frac{s_t}{1-s_t}$ for linear gaussian probability paths)
- we trade exactness for *control* by nudging $\lambda_t = w \cdot a_t \cdot c_t$

Adapted classifier-free guidance

At every flow step t:

1. Estimate the clean state from the current noisy residual: $\hat{x}_t = \hat{x}^{\text{det}} + \hat{r}_t$
2. Score it with the masked loss: $L(\hat{x}_t)$
3. Take the gradient w.r.t. the noisy state: $g_t = \nabla_{z^t} L(\hat{x}_t)$
4. Modify the velocity: $\tilde{u}_t = u_t - \lambda_t \, g_t$ with $\lambda_t = w \cdot a_t \cdot c_t$
5. Take Euler step as usual: $z^{t+1} = z^t + h_t \, \tilde{u}_t$

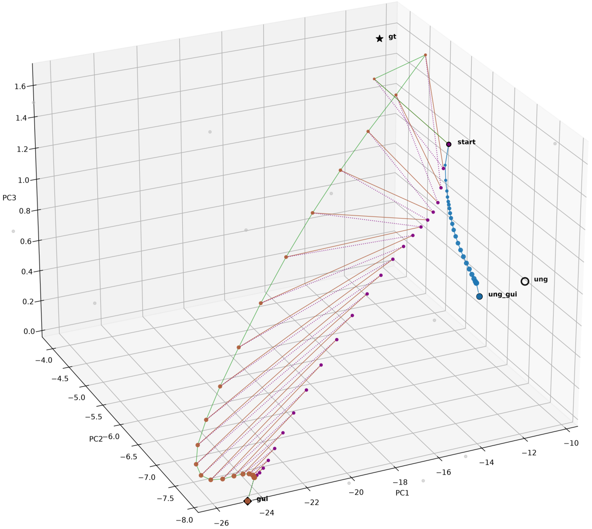


Two design choices define the method:

- **Masked loss** — what to push and where
- **Guidance schedule** — how hard and when

Adapted classifier-free guidance

A guided sampling trajectory in PCA latent-space:



Loss

$$M(x) \;=\; \sum m \odot x \qquad \text{masked average}$$

$$L \;=\; \Big(\, \underbrace{M(\hat{x}_t)}_{\text{current}} - \underbrace{(1+\delta)\, M(x^{\text{ref}})}_{\text{target}} \,\Big)^2 \;=\; e_t^2$$

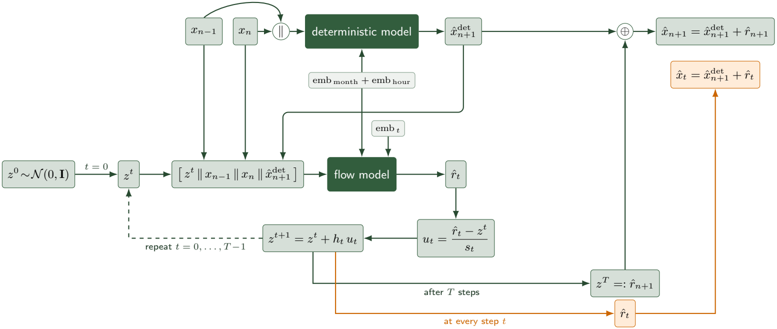
$\hat{x}_t$	clean-state estimate at flow step t (where we are)
x^{ref}	unguided forecast (or ground truth)
δ	target shift — the imposed change
e_t	the **gap**: signed difference, driven to zero

Gradient flows *through the model* into z_t: the network Jacobian spreads the correction over the whole field.

Loss: where $\hat{x}_t$ comes from

Introduction
Designing guidance
Experiments
Work in progress

At every flow step t, the Euler state already yields an estimate of the final residual — and with it the clean state the loss scores (in orange):



s_t: noise level $\quad h_t$: Euler step size

Guidance schedule

Applied strength factorizes: $\lambda_t = w \cdot a_t$

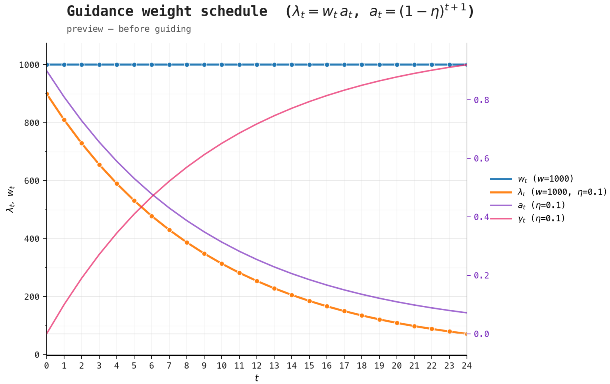
- w — guidance scale: *how hard* to push
- a_t — guidance profile: *when* to push

Gap-closing adaptive profile: conceptually close a fraction η of the remaining gap each step, $e_{t+1} = (1-\eta)\, e_t$, such that:

$$\begin{aligned} \text{gap}: \; & e_t = (1-\eta)^{t+1}\, e_0 \quad \text{with } e_0 = \eta \\ \text{profile}: \; & a_t = (1-\eta)^{t+1} = e_t / e_0 \\ \text{closed gap}: \; & \gamma_t = 1 - (1-\eta)^t \end{aligned}$$

- $\eta = 1$: whole gap in one step; smaller η: spread over the flow
- Early steps do the heavy lifting; late steps refine.
- Different implications depending on guidance method
- This is just some heuristic I defined

Guidance @ flow time: profile & gap



Schedule a_t falls while the closed fraction $\gamma_t = 1 - (1-\eta)^t$ rises; applied strength $\lambda_t = w\, a_t$ on the left axis.

Guidance methods

Both steer with $\lambda_t = w \cdot a_t$ on the same profile $a_t = (1 - \eta)^{t+1}$.

They differ only in *how the strength is set*:

- **CLOSE-GAP** — no w: at each step pick the *exact* λ_t that lands the gap on its prescribed path. Self-correcting, one pass.
- **OPTIMIZE-STRENGTH** — one global scale w, *tuned* against the final loss by an outer optimization; the profile a_t is fixed.

CLOSE-GAP

No scale w — at every step choose λ_t to close the gap *exactly*.

- Aim at the prescribed remaining gap: $e_t^{\text{target}} = (1-\eta)^{t+1}\, e_0$
- Land there in one move along $g_t = \nabla_{z_t} L$:

$$\lambda_t = \frac{2\, e_t\, \left(e_t - e_t^{\text{target}}\right)}{h_t\, \|g_t\|^2}$$

- Anchored to the *measured* gap e_t each step
  $\rightarrow$ drift (model momentum) self-corrects at the next step

Trade-off: follows the profile a_t exactly, but generates a highly oscillatory behaviour, unless a_t is tuned to minimize the oscillations.

Cost: a single guided flow with a gradient pass per step $\sim 2T$

OPTIMIZE-STRENGTH

Fix the profile a_t; tune scale w against final loss at step T:

- Run the whole guided flow at constant w: $\lambda_t = w \cdot a_t$
- Compute final loss as function of w:

$$L_{\mathsf{final}}(w) = L\big(x_T^{\mathsf{gui}}(w)\big) = \Big(M\big(x_T^{\mathsf{gui}}(w)\big) - (1+\delta)\, M(x^{\mathsf{ref}})\Big)^2$$

- Iteratively solve $\frac{dL_{\mathsf{final}}}{dw} = 0$ by the secant method
- Stop when the final loss is closed, then rerun once at w^*
- Note: the realized path does *not* follow the prescribed a_t

Trade-off: reaches the target with less oscillatory behaviour at the cost of the outer optimization over w

Cost: K full flow evaluations with a gradient pass per step $\sim K \cdot 2T$.

Notation (again)

$n = 1, \dots, N$	weather steps (24 h each)
$t = 0, \dots, T$	flow steps within one weather step
x_n, $\hat{x}^{\text{det}}$	weather state, deterministic prediction
z^t, u_t	noisy residual latent, its velocity
s_t, h_t	noise level, Euler step size
$\hat{x}_t$	clean-state estimate at flow step t
m, $M(x)$	mask, masked average $\sum m \odot x$
δ_n, x^{ref}	prescribed change, reference state
L, e_t	masked loss, gap to the target
$g_t = \nabla_{z^t} L$	guidance gradient
$\lambda_t = w \cdot a_t$	guidance schedule: scale $\times$ profile
η	gap-closing rate per flow step

Experiments

Setup parameters

Parameter	Meaning
Start time	input-state timestamp at $n = 0$
M	number of ensemble members
N	number of rollout steps
T	number of flow steps per rollout step
Reference	unguided forecast
$\{\delta_n\}_{n=1}^{N}$	prescribed target change trajectory
a_t	guidance profile
Mask	target region and variable
Mask mode	BBOX or Gaussian mask

Experiments

- Compare guidance methods and η
- Compare η for OPTIMIZE-STRENGTH mode
- Compare change levels $\{\delta_n\}_{n=1}^{N}$
- Compare BBOX and Gaussian mask
- Compare different masks and start times
- Compare different profiles a_t

Evaluation

- Because we are targeting a specific mask, our analysis should consider the mask in particular.
- In fact the guided trajectory has now lost its global spatial coherence (this we specifically aknoledge, and is part of the practical design of the algorithm), but we hope that inside the mask (more specifically inside the box, or near to the center of the mask) things are locally coherent.
- This is in contrast to methodologies that try to achieve the same change but with global coherent trajectories.

Qualitative analysis: guiding questions

Simple:

- Was the guidance target realized?
- How does the guidance look like over the rollout?
- How does the guidance look over the flow?
    - Is change localized, does it spread outside the mask?
    - What happens to the gradient during the flow?
    - Does the prescribed change look reasonable?
    - Does the guidance result surprise you?
- Which variables are influenced from the guidance and how?

Harder:

- What happens when we stop guiding?
- At which level of guidance things start to fall apart?
- What do you expect the optimal guidance schedule to look like?

Quantitative analysis

- cross variable changes
- latent trajectories
- guidance ping-pong analysis

Limitations

- **Fixed mask** — the region is placed by hand and held still: well suited to stationary heatwaves, but not to cyclones or other moving events.
- **Hand-designed target** — the mask and the change trajectory are ours to design, and the loss says nothing about their physical plausibility; realism must be validated separately — for the inputs and the generated states alike.
- **Indirect control** — loss-based guidance nudges the flow toward a target rather than sampling the model's own transition, so the learned dynamics are respected only indirectly.

$\rightarrow$ Sampling-based guidance would address all three at once — no hand-placed mask, no prescribed trajectory, no separate realism check — at the cost of heavier sampling and its own trade-offs.

Contributions

- A **new approach** to guide in the weather domain
- A statement about the **best weight schedule** — given a proper criterion, the whole λ_t schedule should be optimizable at once
- A more comprehensive **analysis of physical realism**, with insights into its different perspectives

Work in progress

TODOs

- **Results**: define results of interest and prepare charts + tables + final experiments
- **Real heatwaves** as targets: Canada 2021, this year's over Europe
- A **baseline**: shift the state directly instead of a gradient update (universal guidance)
- A **distributional, climatology-based** definition of the change trajectory
- **Mask-size experiments**: vary the mask size around the same epicenter

Future work

- Try out **Monte Carlo** approach to estimate $p(y|z_t)$
- Apply the method to **GenCast**
- Estimate the **likelihood** of the guided states under the model